🎓 University Academic & Research Management System (UARMS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

📌 Executive Summary

The University Academic & Research Management System (UARMS) is a structured academic enterprise simulation platform developed entirely in ANSI C.

The system integrates core university operations including:

• Secure administrator authentication
• Student record management
• Faculty management
• Course administration
• Student enrollment tracking
• Research record management
• Grade analytics
• Timestamp-based audit logging

UARMS models centralized academic workflow using deterministic structured programming and binary file persistence without reliance on external databases.

🏗️ System Architecture

UARMS follows a layered logical structure:

• Authentication Layer
• Student Management Module
• Faculty Management Module
• Course Management Module
• Enrollment Module
• Research Management Module
• Grade Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Although implemented within a single source file, functional responsibilities remain logically separated.

🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `university_users.dat`

Default Credentials:

Username: admin
Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

👨‍🎓 Student Management

Each student record contains:

• Student ID
• Name
• Department
• CGPA

Functional Capabilities:

• Add student
• View student records

Storage File: `students.dat`

👨‍🏫 Faculty Management

Each faculty record contains:

• Faculty ID
• Name
• Department
• Salary

Functional Capabilities:

• Add faculty
• View faculty records

Storage File: `faculty.dat`

📚 Course Management

Each course record contains:

• Course ID
• Course Name
• Credit

Functional Capabilities:

• Add course
• View course records

Storage File: `courses.dat`

📝 Enrollment Module

Each enrollment record contains:

• Student ID
• Course ID
• Grade

Functional Capabilities:

• Enroll student into course
• Store grade information

Storage File: `enrollments.dat`

🔬 Research Management

Each research record contains:

• Research ID
• Title
• Researcher
• Domain

Functional Capabilities:

• Add research record

Storage File: `research.dat`

📊 Grade Analytics Engine

The system performs statistical analysis on student grades.

Metrics Computed:

• Mean (Average Grade)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Grades / N
Variance = Σ (Grade − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Evaluate academic performance distribution
• Measure grade variability
• Analyze academic consistency

📝 Audit Logging

All major system activities are recorded in:

```
university_logs.txt
```

Logged Events:

• Admin login
• Student addition
• Faculty addition
• Course addition
• Enrollment activity
• Research addition
• Grade analytics execution
• System exit

Each log entry includes timestamp metadata.

💾 Data Storage Design

Binary files used:

File	Purpose
university_users.dat	Authentication records
students.dat	Student records
faculty.dat	Faculty records
courses.dat	Course records
enrollments.dat	Enrollment records
research.dat	Research records
university_logs.txt	Audit logs

Binary storage ensures:

• Structured record alignment
• Faster read/write operations
• Reduced storage overhead
• Persistent academic data integrity

⚙️ Compilation & Execution

Compile:

```
gcc university_management_system.c -o university -lm
```

Run:

```
./university
```

🧠 Engineering Highlights

• ANSI C structured programming
• Academic workflow abstraction
• File-based database simulation
• Statistical grade analysis
• Deterministic memory usage
• Timestamp-based traceability
• Modular logical segmentation

🎯 Project Significance

UARMS demonstrates the capability to design and implement a centralized academic management system using low-level systems programming.

It reflects:

• Academic operational modeling
• Structured data persistence
• Analytical computation integration
• Secure authentication handling
• Enterprise-grade logical organization

🏁 Conclusion

The University Academic & Research Management System (UARMS) is a complete academic enterprise simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Student Management
Faculty Management
Course Administration
Enrollment Tracking
Research Management
Grade Analytics
Audit Logging

All within a structured and deterministic runtime architecture.